

● INTRODUCTION AND BACKGROUND

Motivation

Depression affects **332 million people globally**. A hallmark of MDD is **psychomotor retardation** reduced speed and amplitude of voluntary movement, 360-degree VR provides a naturalistic context to measure this unobtrusively; depressed individuals are predicted to **rotate their head less** due to blunted affect and diminished motivation.

● PRIOR LITERATURE

Srivastava et al. (2025) In 360-deg VR, moderately-to-severely depressed individuals (PHQ-9 ≥ 10) showed substantially lower head-rotation speed after covarying for anxiety (ANCOVA: $p < 0.001$, $\eta^2 = .295$). Primary measure: SDY (SD of yaw angular position).

Bennabi et al. (2013) Psychomotor retardation in depression confirmed as core observable biomarker of MDD.

Kroenke et al. (2001) PHQ-9 showed strong validity and can be treated as a continuous depression-severity predictor in non-clinical samples.

● DATASET AND PARTICIPANTS

Sample Description (N = 40 postgraduate students)

	40	22.8	8	32	40%
Total participants		Mean age SD = 1.8	Depressed PHQ-9 ≥ 10	Non-dep PHQ-9 < 10	Prior VR experience

MEASURE	MEAN	SD	RANGE
PHQ-9 (Depression)	6.02	4.63	0-18
GAD-7 (Anxiety)	5.00	4.31	0-18
STAI-T (Trait Anxiety)	45.0	14.63	21-73
PANAS Positive Affect (pre)	34.52	7.33	17-47
PANAS Negative Affect (pre)	12.70	3.52	10-26

36 male, 4 female · r(PHQ-9, GAD-7) = .58, $p < .001$ anxiety must be controlled

● VR STIMULI - FIVE 360 DEGREE VIDEOS

ID	VIDEO	VALENCE	AROUSAL	PRESENCE	SPEED (DEG/S)
V1	Abandoned Buildings	5.20	4.60	29.60	39.08 (highest)
V2	Beach	6.53	4.65	27.25	32.75
V3	Campus	6.15	4.33	26.95	34.87
V4	Horror (Nun)	3.67	5.95	19.85	24.17 (lowest)
V5	Tahiti Surf	7.12	6.50	28.02	31.03

Head-tracking: Mean speed (deg/s), Yaw speed, SD rotation speed, Angular range, SDY (SD of yaw - Srivastava primary measure).

● METHODOLOGY - ANALYSIS PIPELINE

10-Step Statistical Pipeline

1 RM-ANOVA + Mauchly Sphericity Test

One-way within-subjects ANOVA across 5 videos. Mauchly $W = 0.74$, $p = .257$ (sphericity met). Post-hoc: paired t-tests, Holm correction.

2 Kruskal-Wallis + Pairwise Mann-Whitney U

Non-parametric video effect. 10 pairwise M-W U tests, Bonferroni-corrected as robustness check.

3 Friedman's ANOVA (Non-parametric RM)

Ranks within participants; no distributional assumptions. Kendall's W effect size.

4 Simple Linear Regression per Video

PHQ-9 (continuous) predicting speed per video. Holm + BH-FDR multiple comparison correction.

5 ANCOVA with Anxiety Covariates

Speed ~ PHQ-9(z) + GAD-7(z) + STAI-T(z), z-standardised. Type II partitioning. Replicates Srivastava et al.

6 Linear Mixed Model (LMM)

Full repeated-measures. Random participant intercept (REML). ML for AIC comparison. ICC for individual clustering.

7 MCC: 30 Tests (5 videos x 6 measures)

Added SDY (Srivastava key measure). Bonferroni, Holm, BH-FDR compared for transparency.

8 Post-hoc Power Analysis

Power at original effect ($\eta^2 = .295$, $d = 1.29$). MDE at 80% power.

9 PHQ-9 to Subjective VR Ratings

LMMs for valence, arousal, presence. Wilcoxon within-group PANAS change. Mann-Whitney between-group.

10 Moderation: Multiple Linear Regression

Speed ~ PHQ(z) + Presence(z) + PHQ(z) x Presence(z). Tests if VR immersion moderates depression to head-tracking link.

● RESEARCH QUESTIONS

Four Research Questions Tested

RQ1 - VIDEO MAIN EFFECT
Does head-rotation speed differ significantly across the five 360-degree videos and which pairs of videos drive that effect?

Supported - $F(4,156)=34.58$, $p<.001$, $\eta^2=.181$

RQ3 - SUBJECTIVE VR AND PANAS
Does depression moderate the subjective experience of VR (valence, arousal, presence ratings) or PANAS affect change?

Partial - Arousal $b=-0.44$, $p=.012$; PA drop $d=-1.54$, $p=.008$

RQ2 - PHQ-9 AS CONTINUOUS PREDICTOR
Does PHQ-9 score (continuous predictor) predict mean head-rotation speed, before and after covarying for anxiety (GAD-7, STAI-T)?

Not Supported - $R\text{-sq}=.02$; adj. $R\text{-sq}$ negative; LMM $b=-0.55$, $p=.71$

RQ4 - PRESENCE MODERATION
Does VR presence moderate the relationship between depression and head-movement?

Not Supported - PHQ x Presence interaction $p=.27$

Supporting analyses: MCC (30 tests): 0/30 survive Bonferroni/Holm/BH-FDR) and power analysis (89.1% at $d=1.29$).

● RESULTS - RQ1: VIDEO MAIN EFFECT ON HEAD-ROTATION SPEED

Head-Rotation Speed Strongly Varies by Video Content

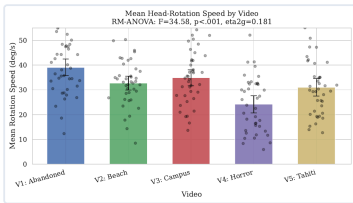


Fig 1. Mean speed by video. RM-ANOVA: $F(4,156)=34.58$, $p<.001$, $\eta^2=.181$. Mauchly $W=0.74$, $p=.257$ (sphericity met).

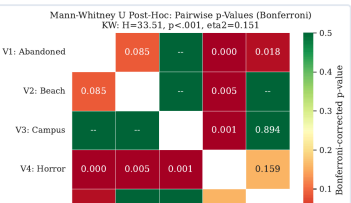
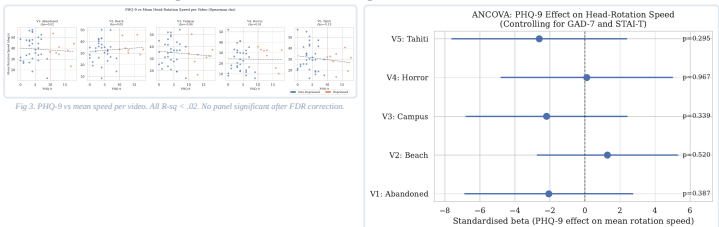


Fig 2. Bonferroni-corrected Mann-Whitney U post-hoc. 4/10 pairs significant -- all involving V4 (Horror).

F=34.58 RM-ANOVA $F(4,156)$ $p < .001$	.181 η^2 -sq-g 18% variance	70.96 Friedman chi-sq(4) $p<.001$, $W=0.444$	33.51 KW H(4) $p<.001$, η^2 -sq-g=.151
Post-hoc (Holm): V4 Horror significantly lower than all others (all $p < .001$, $d > 0.84$). V1 Abandoned highest (39.08 deg/s). V2 vs V3 not significant ($p = .195$). Three convergent tests (RM-ANOVA, Kruskal-Wallis, Friedman) confirm: video content is the dominant driver.			

● RESULTS - RQ2: PHQ-9 DOES NOT PREDICT HEAD-ROTATION SPEED (REGRESSION, ANCOVA, LMM)

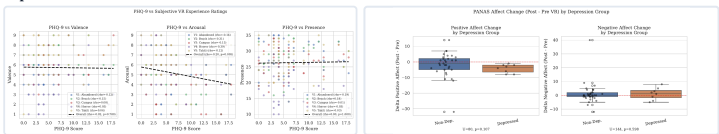
No Association Between Depression and Head-Rotation Speed



All PHQ-9 slopes near zero ($|\beta| < 0.35$, all $R\text{-sq} < .02$). ANCOVA: all adjusted $R\text{-sq}$ negative (predictors explain less than null model). PHQ-9 shows no unique relationship with head-tracking even after controlling for anxiety (all $p > .29$, 0/5 survive FDR).

● RESULTS - RQ3: SUBJECTIVE VR EXPERIENCE AND PANAS AFFECT CHANGE

Depression Blunts Arousal and Reduces Positive Affect



b=-0.44 Arousal $p=.012$	b=-0.03 Valence $p=.302$ (ns)	d=-1.54 PA drop $p=.008$	p>.27 RQ4: Presence interaction (ns)
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PHQ-9 predicts lower arousal (anhedonia). Depressed group: large PA decrease post-VR ($d=-1.54$, $p=.008$); non-depressed: no change ($p=.247$). Affective measures appear more sensitive than motor measures in sub-clinical samples. **RQ4: VR presence did not moderate the depression-headtracking link** (all interaction $p>.27$).

● RESULTS - RQ2 (CONT.): LINEAR MIXED MODEL : FULL REPEATED-MEASURES ANALYSIS

Individual Differences Dominate: ICC = 0.694

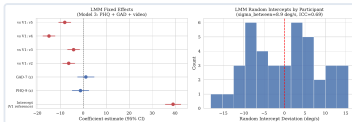


Fig 5. LMM fixed effects (left) + random intercept distribution (right). SD 8.9 degrees between vs 5.9 within.

MODEL	B-PHQ	P	AIC
M2: PHQ + video	-0.55	.710	1388.2
M3: PHQ+GAD+video	-1.20	.508	1389.8
M4: PHQ+video	---	---	1391.9

0.694

ICC: 69% of variance from individual style

-0.55

b-PHQ (M2)
 $p=.71$ (ns)

ICC = 0.694: 69% of total variance is stable individual scanning style. This noise floor buries the depression signal. Random intercept in LMM controls for this; PHQ-9 still non-significant.

● ADDITIONAL ANALYSES - MCC (30 TESTS) AND POST-HOC POWER ANALYSIS

0/30 Tests Survive Any Correction · Power = 89%

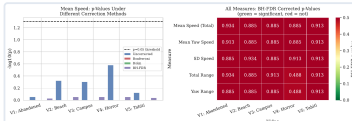


Fig 6. $-\log_{10}(p)$ for 4 MCC methods + BH-FDR heatmap (6 measures incl. SDY). Zero bars exceed alpha=.05 after any correction.

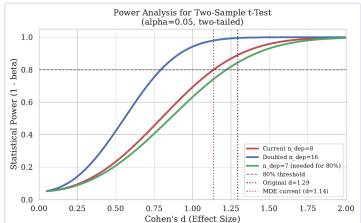


Fig 7. Power curves: current design achieves 80% at original $d=1.29$. MDE=1.14 at 80% power.

CORRECTION METHOD	SIG / 30 TESTS
Uncorrected	3 (chance-level)
Bonferroni	0 / 30
Holm	0 / 30
BH-FDR	0 / 30

SDY (Srivastava key measure): V4 $p=.025$ uncorrected to $p=.74$ after Bonferroni. Other videos $p > .70$. Primary measure does not replicate.

● CONCLUSIONS - ANSWERING THE FOUR RESEARCH QUESTIONS

Summary of Key Findings (Mapped to Report 2 RQs)

- RQ1 - Video effect:** Supported. RM-ANOVA: $F(4,156)=34.58$, $p<.001$, $\eta^2=.181$. Horror V4 lowest (freezing/hypervigilance response). Kruskal-Wallis ($H=33.51$) and Friedman (chi-sq=70.96, $W=0.444$) converge. V4 significantly lower than all others (all Holm-corrected $p<.001$, $d>0.84$). Video content is the dominant driver.
- RQ2 - PHQ-9 as continuous predictor:** Not Supported. Simple regression: all $R\text{-sq} < .02$, all $p > .38$ (0/5 survive FDR). ANCOVA (PHQ+GAD+STAI): all adjusted $R\text{-sq}$ negative, PHQ-9 non-significant in all 5 videos (all $p > .29$). LMM: $\beta=-0.55$, $SE=1.47$, $p=.71$. SDY (Srivastava primary measure): 4/5 videos $p > .70$; V4 $p=.025$ uncorrected to $p=.74$ after Bonferroni. 0/30 tests survive any correction.
- RQ3 - Subjective VR experience and PANAS:** Partially Supported. Arousal: PHQ-9 significantly predicts lower arousal ratings ($\beta=-0.44$, $p=.012$; anhedonia consistent with DSM-5 A2). Valence and presence: not significant ($p=.332$, $p=.341$). PANAS: Depressed group showed a large, significant decrease in positive affect post-VR ($d=-1.54$, $p=.008$); non-depressed showed no change ($p=.247$).
- RQ4 - Presence moderation:** Not Supported. Multiple linear regression with PHQ(z) x Presence(z) interaction term: all interaction $p > .27$ across all 5 videos. VR immersion does not buffer or amplify any latent depression-related scanning differences. Presence rules out as a confound (depressed participants did not feel less immersed; presence LMM: $p=.841$).
- Why did replication fail?** ICC=0.694 - 69% of variance from stable individual scanning style buries any depression signal. More critically, 80% of the sample scored below PHQ-9=10 (sub-clinical).

● LIMITATIONS

Small groups: n-dep=8. Power for $d=0.5$ drops to 23%.	Sub-clinical range: 80% below PHQ-9=10. Signal may only emerge in severe.
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● KEY REFERENCES

- Srivastava et al. (2025)** - What Do Head Scans Reveal About Depression? Insights from 360 Psychomotor Assessment
- Bennabi et al. (2013)** - Psychomotor retardation in depression: a systematic review of diagnostic, pathophysiological, and therapeutic implications
- Kroenke, Spitzer and Williams (2001)** - The PHQ-9: Validity of a brief depression severity measure
- Spitzer et al. (2006)** - A brief measure for assessing generalized anxiety disorder: The GAD-7
- Watson, Clark and Tellegen (1988)** - Development and validation of brief measures of positive and negative affect: The PANAS scales
- Spielberger et al. (1983)** - Manual for the State-Trait Anxiety Inventory
- Kourtesis et al. (2019)** - (VR)ISE Validation of the Virtual Reality Neuroscience Questionnaire: Maximum duration of immersive virtual reality sessions without the presence of perturbation